
Domain Adaptation and Generalization Across Vision and Vision-Language Models

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Abstract

We evaluate three strategies for robustness across unseen domains: i) domain adaptation, ii) domain generalization, iii) prompt tuning in vision-language models. In domain adaptation, CDAN and DAN achieve comparable performance, while DANN underperforms due to over-alignment and loss of class separability. ERM remains a strong DG baseline, often matching or surpassing IRM and GroupDRO, while ERM and SAM preserve discriminative ability across seen and unseen domains. Prompt tuning shows trade-offs between specialization and generalization, with zero-shot CLIP often matching fine-tuned models. These findings highlight the balance between invariance, discriminability, and prompt-induced specialization for cross-domain robustness. Code and reproducibility material is available at: [code](#).

1. Introduction

Generalizing to unseen domains is a key challenge for deep learning models. Two major paradigms address this issue: Domain Adaptation (DA) and Domain Generalization (DG). While DA assumes access to unlabeled samples from the target domain during training, DG seeks to learn models that perform well on entirely unseen domains (Gulrajani & Lopez-Paz, 2020; Wang et al., 2022).

We first explore domain adaptation strategies to analyze how explicitly minimizing divergence between domains influences transfer performance. Using the PACS dataset (Yu et al., 2022) with *photo* as the labeled source and *art painting* as the unlabeled target, we evaluate Empirical Risk Minimization (ERM), Deep Adaptation Network (DAN), Domain-Adversarial Neural Network (DANN), Conditional Domain-Adversarial Network (CDAN), and pseudo-label self-training. CDAN and DAN achieve comparable perfor-

mance, consistently outperforming DANN, which suffers from reduced class separability due to over-alignment.

Next, we evaluate DG algorithms, Invariant Risk Minimization (IRM) (Arjovsky et al., 2020), Group Distributionally Robust Optimization (GroupDRO) (Bai et al., 2025), and Sharpness-Aware Minimization (SAM) (Foret et al., 2021) against ERM as a baseline. Experiments on PACS with a ResNet-50 backbone (He et al., 2015) reveal that ERM remains highly competitive, corroborating findings that DG methods do not consistently outperform strong baselines (Gulrajani & Lopez-Paz, 2020). While invariance-based methods can stabilize cross-domain performance, discriminative features learned through ERM are often sufficient.

A complementary focus is prompt tuning in vision-language models like CLIP. Although CLIP’s contrastive pretraining enables remarkable zero-shot transfer, deploying it in specific applications often requires adapting prompts without modifying the frozen backbone. We investigate how prompt optimization affects open-set recognition, cross-domain adaptation, and multi-domain stability. Comparing zero-shot CLIP with fine-tuned linear probes reveals a trade-off: adaptation improves seen-domain accuracy but can degrade performance on unseen domains. We further analyze pseudo-labeling, entropy minimization, and gradient conflicts across domains, showing that effective adaptation requires controlled prompt drift and explicit alignment mechanisms to preserve CLIP’s intrinsic semantic structure.

2. Methodology

2.1. Domain Adaptation

2.1.1. EXPERIMENTAL SETUP

We use the PACS (Yu et al., 2022) dataset to evaluate various domain adaptation strategies. The dataset comprises four different domains: *art painting*, *photo*, *cartoon*, and *sketch*. Each domain contains seven classes: *dog*, *elephant*, *giraffe*, *guitar*, *horse*, *house*, and *person*. In our setup, the *art painting* domain is kept as the domain without class labels, while *photo* is kept as the in-domain dataset with both (x, y) present.

We use a ResNet-50 (He et al., 2015) as the feature extractor

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Model	Setting	Train Loss	Val Photo Acc (%)	Val Art Acc (%)
ERM	Baseline	0.0201	99.70	63.00*
DAN	No Shift	0.0172	98.50	75.31
DAN	Label Removed	0.0243	99.10	77.27
DAN	Undersampled	0.1792	99.40	78.75
DANN	No Shift	0.1465	98.80	69.68
DANN	Label Removed	0.1393	99.40	72.16
DANN	Undersampled	0.1533	99.10	74.22
CDAN	No Shift	0.1328	99.40	75.06
CDAN	Label Removed	0.1646	97.90	77.56
CDAN	Undersampled	0.0698	99.70	74.79
Pseudo-labels	Baseline	2.7842	97.90	46.21

Table 1. Best validation results for each domain adaptation strategy under different experimental settings (evaluated on the entire *art painting* set).

in all our experiments. In order to adapt the model for our experiments, we add a classifier with seven output nodes, matching the number of classes in our dataset. Additional architectural changes are introduced depending on the specific domain adaptation strategy, which are detailed in the subsequent sections.

All models are trained for 20 epochs using a learning rate of 1×10^{-4} and a batch size of 64. Both the in-domain and out-of-domain datasets are split into training and evaluation sets with an 80/20 ratio, unless otherwise specified in the following sections.

We conduct three experimental settings to analyze model behavior under different types of distribution shifts: (1) a baseline setup with no artificial shifts, (2) a *label-shift* setup where the *giraffe* class is entirely removed from the target *art painting* domain, and (3) a *rare-class* setup where the *giraffe* class in the target domain is downsampled to only five instances. These settings allow us to evaluate the robustness of each adaptation method under varying target-domain conditions.

2.1.2. TRAINING STRATEGIES

We evaluate the performance of our model on both domains under the following domain adaptation strategies: (i) Empirical Risk Minimization (ERM), (ii) Domain Adversarial Neural Network (DANN), (iii) Domain Alignment Network (DAN), (iv) Conditional Domain Adversarial Network (CDAN), and (v) self-training on the target domain (Pseudo-labels). All networks are implemented from scratch without relying on any external libraries.

ERM: The ERM model serves as the baseline for evaluat-

ing all other domain adaptation strategies. The model is trained with a vanilla cross entropy loss function on the *photo* domain. It never sees the *art painting* domain during training. The ERM model is trained and evaluated under baseline (no label shift) scenario.

DANN: We modify our baseline model by incorporating a domain discriminator that uses the feature representations extracted by the ResNet-50 backbone. The discriminator consists of three fully connected layers with two ReLU activation functions and dropout applied at a rate of 0.5.

We use the cross-entropy loss to optimize the classifier head and the binary cross-entropy loss to train the domain discriminator. The feature extractor is updated using a combination of the classification loss and the weighted domain loss, implemented via a gradient reversal layer (GRL). The weighting coefficient λ is linearly increased from 0 to 0.2 throughout training. The classification loss is computed only on samples from the *photo* (source) domain.

DAN: We train the model using classification loss on samples from the *photo* domain and an additional alignment loss on the features extracted from layer 4 of the ResNet-50 backbone. The alignment loss matches the means of the source and target feature distributions to reduce domain shift between the features extracted.

CDAN: Our CDAN model uses the same architecture as our DAN model. However, in line with the literature, our CDAN model uses the outer product of the feature representation z and the class probabilities p to discriminate between the *photo* and *art painting* domains

Pseudo-labels: Our ERM model is used to generate labels for the *art painting* set. We kept only those samples that

Class	Src Train	Tgt Train	F1 ERM	F1 DANN	F1 CDAN	F1 DAN
dog	152	308/300/305	0.752	0.755/0.713/0.814	0.771/0.859/0.738	0.794/0.853/0.760
elephant	158	205/200/212	0.704	0.748/0.803/0.800	0.746/0.807/0.756	0.772/0.796/0.795
giraffe	144	223/0/3	0.704	0.806/0.000/0.333	0.825/0.000/0.211	0.826/0.000/0.111
guitar	145	152/147/137	0.528	0.732/0.838/0.833	0.800/0.829/0.836	0.723/0.761/0.822
horse	163	166/160/167	0.577	0.600/0.681/0.607	0.635/0.753/0.674	0.634/0.654/0.800
house	221	245/237/246	0.884	0.943/0.966/0.938	0.961/0.975/0.947	0.920/0.919/0.947
person	353	340/367/345	0.322	0.375/0.526/0.614	0.648/0.583/0.682	0.656/0.714/0.828

Table 2. Per-class F1-scores on the target domain (*art_painting*) across three conditions (Normal / Label Removed / Label Undersampled). Each value represents results under the respective setting in order. If it is the same across different conditions, it is only written once.

were predicted by the model with a confidence > 0.5 . The labeled *art_painting* samples are then used to fine-tune the ERM model. This approach is evaluated only under the baseline (no-shift) setting.

2.1.3. EVALUATION METRICS

We evaluate the performance of our models using a combination of quantitative and qualitative metrics:

- (1) We report the classification accuracies of all models on the *photo* and *art_painting* validation sets.
- (2) We report the F1-scores of the three rarest classes in both domains for models trained under the baseline (no-shift) setting.
- (3) We visualize the embeddings extracted from the feature extractor for all experiments except Pseudo-labels.

2.2. Domain Generalization

2.2.1. EXPERIMENTAL SETUP

To evaluate various domain generalization strategies, we use a pretrained ResNet-50 backbone and the PACS dataset as the baseline for all experimentation. In our setup, the Sketch domain is held out as the unseen target domain, while the remaining three (Photo, Art, and Cartoon) are merged and used as the source domains during training. For each source domain, the dataset is divided into an 80/20 train-validation split to ensure a balanced evaluation of model generalization.

To adapt the ResNet for PACS, we replace its final fully connected layer with a classifier matching the dataset’s number of output classes. Furthermore, we make two modifications to the architecture, which make the model more robust for generalization (Gulrajani & Lopez-Paz, 2020): (i) all Batch Normalization layers are frozen, and (ii) a dropout layer with a probability of $p = 0.3$ is added.

2.2.2. TRAINING STRATEGIES

We evaluate the generalization performance of the following training strategies on unseen target domains: i) Empirical Risk Minimization (ERM), ii) Invariant Risk Minimization (IRM), iii) Group Distributionally Robust Optimization (GDRO), and iv) Sharpness Aware Minimization (SAM).

We utilize the `DomainBed` suite to train all models (Gulrajani & Lopez-Paz, 2020). Since SAM is not included in the suite, we manually implemented it following the first-order approximation proposed in the original paper.

Each model is trained for 5,000 steps using a learning rate of 5×10^{-5} and a batch size of 32. However, SAM is trained with a larger learning rate of 1×10^{-3} , which was also done in the original paper. Model checkpoints are saved and evaluated at regular intervals to monitor performance throughout training. All models are trained using the Adam optimizer, except SAM, which employs Stochastic Gradient Descent (SGD) — the optimizer conventionally used for SAM in the literature. Model checkpoints are saved every 1,000 steps, except for GDRO, where checkpoints are created every 500. The entire experimental setup is evaluated with a fixed set and deterministic operations to allow for reproducibility. For all models, we use accuracy as a baseline metric to evaluate performance.

ERM: The ERM model serves as the baseline for evaluating all other domain generalization strategies.

IRM: For IRM (Arjovsky et al., 2020), we evaluate the model under varying penalty strengths $\lambda = \{0.1, 10, 100, 10000\}$ to analyze their effect on invariance and stability. For all these experiments, we use an IRM penalty annealing of 500 iterations, during which the penalty weight is kept at 1.0 to allow for stable training before gradually increasing it to its full value. Owing to IRM’s tendency to collapse, we ensure that all check-pointed models are evaluated throughout training to capture meaningful improvements. To get a measure of cross-domain flatness,

Algorithm	Best Step	Target Sketch (%)	Photo (%)	Art (%)	Cartoon (%)
ERM	3000	72.74	95.11	97.22	98.20
IRM ($\lambda=0.1$)	1000	74.52	93.89	96.58	97.31
IRM ($\lambda=10$)	3000	74.90	95.60	97.86	98.50
IRM ($\lambda=100$)	3000	5.86	23.47	20.94	30.43
IRM ($\lambda=10000$)	1000	7.64	20.54	24.36	31.44
GroupDRO ($\eta=0.1$)	1000	71.59	96.09	97.44	98.50
GroupDRO ($\eta=0.01$)	1000	73.38	95.89	95.94	97.60
GroupDRO ($\eta=0.001$)	2500	66.62	96.33	97.44	97.60
SAM ($\rho=0.01$)	5000	70.96	94.62	96.79	99.40
SAM ($\rho=0.05$)	4000	73.30	94.38	97.22	98.80
SAM ($\rho=0.1$)	5000	72.44	93.89	95.73	98.50

Table 3. Accuracy (%) comparison across training strategies on the PACS dataset. The best stop denotes the checkpoint with the highest validation performance.

we perturb the weights and see the effect on the individual domain losses and accuracies. This is to test whether IRM learns a truly domain-general solution, which would have a broad region of low loss that covers all domains. Furthermore, we also keep a track of the penalty term to see whether the model has collapsed and learned a trivial solution.

GDRO: For GDRO, we experiment with varying multipliers $\eta = \{0.001, 0.01, 0.1\}$. Given GDRO’s reliance on strong regularization, we increase the frequency to check pointing every 500 steps so we can identify optimal performance before it degrades since it requires regularization like early stopping (Bai et al., 2025). We observe the worst group accuracy and also how the weights for each domain change with training.

SAM: The SAM algorithm is implemented using a first-order approximation with $\rho = \{0.05, 0.01, 0.1\}$ (Foret et al., 2021). To assess the flatness of the final solution, we compare the post-training loss values after perturbations of the converged models against the ERM baseline. To do this, we use the converged models and perturb the weights and re-evaluate the training loss to see how it’s affected. Additionally, we visualize the corresponding loss landscape to qualitatively examine solution sharpness and generalization behavior. To analyze the flatness of the optimization landscape, we follow the approach of Li et al. (2018). Two random, orthogonal directions are generated in the model’s parameter space using filter-wise normalization (Li et al., 2018). For each parameter tensor, a random direction is sampled, normalized per filter, and scaled by the norm of the corresponding trained filter to preserve layer-wise scale. These directions are used to define a 2D plane along which the loss function is evaluated.

2.3. CLIP

2.3.1. ZERO-SHOT CLIP AND FINE-TUNED LINEAR PROBE EVALUATION

We used CLIP ViT-B/32 model for zero-shot evaluation. For each domain, prompts followed “a {domain_descriptor} of a {class}”. Images were encoded by CLIP’s frozen vision encoder, and predictions were obtained via cosine similarity ($\times 100$) with text embeddings.

For fine-tuning, a linear probe was trained on frozen CLIP features using a leave-one-domain-out protocol. Training used AdamW (lr 1×10^{-3} , weight decay 1×10^{-2} , batch 256) with cosine annealing over 50 epochs, minimizing cross-entropy loss on the classifier. Convergence occurred within 10 epochs.

Probes were evaluated on all four PACS domains, including the held-out target. We report per-domain, mean, and worst-group accuracies relative to the zero-shot baseline.

2.3.2. PROMPT TUNING FOR DOMAIN ADAPTATION

We explored prompt tuning for domain adaptation, using *photo* as the source and *sketch* as the target domain, representing a substantial appearance gap. All seven classes were used in both domains (1,670 labeled source and 3,929 unlabeled target samples). The zero-shot CLIP baseline with handcrafted prompts was first evaluated, measuring accuracy, mean softmax probability (MSP), and entropy as uncertainty metrics.

A lightweight CoOp-style learnable prompt module was implemented, comprising four trainable context vectors (initialized with std = 0.02). These vectors were averaged, added to the base zero-shot text embeddings, and L2-normalized. Optimization employed AdamW (learning rate 5×10^{-4} , weight decay 1×10^{-3} , gradient clipping at norm 1.0) for 10–20 epochs.

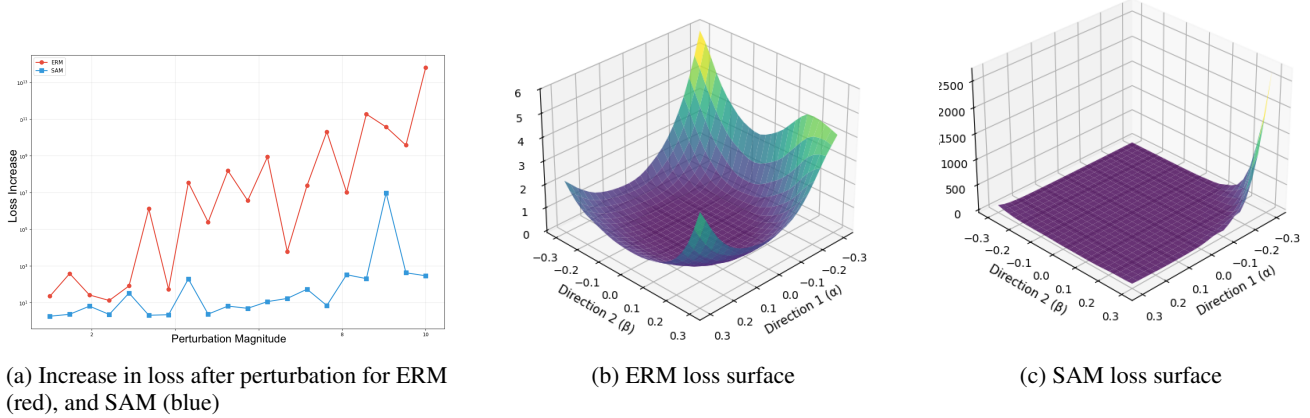


Figure 1. Comparison of loss landscape characteristics across training strategies: (a) change in loss under weight perturbation, (b) ERM loss surface, and (c) SAM loss surface.

Three training strategies were compared: (1) *Source-only supervision*: cross-entropy loss on labeled source data with drift regularization. (2) *Source + Target Entropy Minimization*: combined supervised source loss with an unsupervised entropy term encouraging confident predictions in low-density regions:

$$L = L_{CE}(\text{source}) + \lambda_{\text{ent}} H(\text{target}) + \lambda_{\text{reg}}(1 - \cos_{\text{sim}}),$$

where $\lambda_{\text{ent}} = 0.3$ and $\lambda_{\text{reg}} = 0.1$.

(3) *Source + Target Pseudo-Labeling*: self-training using high-confidence pseudo-labels ($p_{\text{max}} > 0.9$)

Post-training analyses included: (1) target-domain accuracy comparison with zero-shot CLIP, (2) prompt drift via cosine similarity, (3) MSP-entropy analysis for calibration assessment, and (4) correlation between drift magnitude and adaptation performance to examine balance between invariance and specialization.

2.3.3. GRADIENT CONFLICT ANALYSIS IN MULTI-DOMAIN PROMPT LEARNING

To investigate inter-domain gradient conflicts during multi-domain prompt tuning, we selected *art_painting* and *photo* domains, which exhibit strong stylistic contrast but identical class sets. Using the same CoOp-style architecture, only prompt parameters were optimized while the CLIP backbone remained frozen. Prompts were initialized with $\text{std} = 0.02$ and trained using Adam (learning rate 1×10^{-4}) for 30 epochs. The total loss combined both domain objectives.

At epochs 0 to 30, we computed gradients for each domain batch. Flattened gradients g_{art} and g_{photo} enabled geometric analysis via: (1) cosine similarity, (2) gradient norms, and (3) gradient angle

We categorized alignment as follows: $\cos(\theta) > 0.5$ (strong alignment, $\theta < 60^\circ$), $0 < \cos(\theta) \leq 0.5$ (partial, $60^\circ\text{--}90^\circ$),

$-0.5 < \cos(\theta) \leq 0$ (moderate conflict, $90^\circ\text{--}120^\circ$), and $\cos(\theta) \leq -0.5$ (severe conflict, $\theta > 120^\circ$). We correlated these metrics with per-domain loss curves, identifying epochs where loss divergence coincided with severe conflicts and quantifying their impact on final accuracy.

2.3.4. PROMPT TUNING AND OPEN-SET RECOGNITION

We evaluated the impact of prompt tuning on open-set recognition using *sketch* as the source and *photo* as the target domain. Of the seven PACS classes, five (dog, elephant, giraffe, guitar, horse) were used as *seen* classes, while the remaining two (house, person) served as *unseen* classes. The seen subset was divided into 80% training and 20% validation, with 240 samples for unseen evaluation.

A zero-shot baseline was first established by evaluating CLIP model on both seen and unseen classes. Learnable prompts were then trained on seen classes to adapt CLIP’s text representations to the sketch domain. The prompt module comprised 16 context tokens processed by a small MLP and added to the base text embeddings. No unseen or target-domain samples were used during training.

Post-training evaluation included: (1) *Open-set recognition*: assessing seen-class validation and unseen-class performance using hybrid text features (tuned for seen, zero-shot for unseen). (2) *Confidence calibration*: analyzing MSP and entropy of unseen samples under a closed-set classifier trained on seen classes. (3) *Prompt drift*: computing cosine similarity between tuned and zero-shot embeddings. (4) *Few-shot domain adaptation*: retraining prompts on the target (photo) domain using five samples per class for 15 epochs and measuring cross-domain cosine similarity between source and target prompts to assess domain invariance.

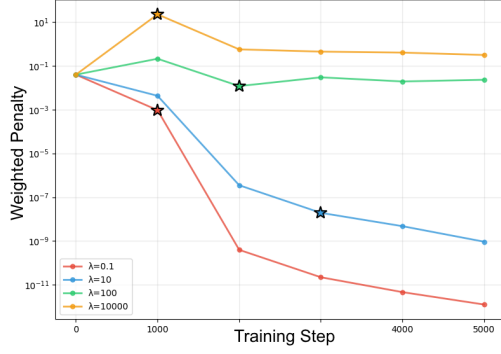


Figure 2. IRM weighted penalty values during training. The stars indicate the models selected based on the best validation criterion.

3. Results

3.1. Domain Adaptation

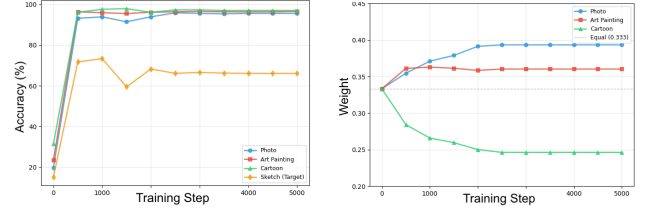
3.1.1. QUANTITATIVE RESULTS

We report the training loss, validation accuracy on the source domain, and validation accuracy on the target domain in Table 1. On average, all three domain adaptation strategies achieve target-domain accuracies that are approximately 10–15% higher than the baseline ERM model, which attains 63% on the target set. Moreover, all three models perform the worst in the ‘No Shift’ experiment achieving an accuracy of 2–3% lower as compared to ‘Label Removed’ and ‘UnderSampled’ experiments. The only exception to this is the CDAN which performs slightly worse in ‘UnderSampled’ experiment. A key outlier in our experiments is the model fine tuned on the Pseudo-labels model as it performs significantly worse of the validation set of Art attaining an accuracy of 46% which is even lower than the ERM.

In order to quantify the effect of differing class frequencies on domain adaptation performance, we report the per-class F1-scores for all models under three controlled target-domain conditions. Table 2 presents the *normal* (*no-shift*) setting, where all seven classes are present with their natural frequencies. This serves as the baseline distribution for comparison. Table 2 also reports results under the *label-removed* condition, where the *giraffe* class is completely omitted from the target domain to simulate a missing-class scenario. Finally, the same table includes the *label-undersampled* setting, where *giraffe* is severely under-represented, illustrating the impact of rare-class scarcity on adaptation.

Our results yield two key insights. First, there is no consistent correlation between class frequency and adaptation performance. For instance, in the *normal* setting, the *horse* class (166 target samples) achieves an F1 comparable to or higher than the *elephant* class (205 samples), while the *person* class, despite having the largest number of target samples, shows the lowest F1 across all models. This suggests

that domain shift and intra-class variability have a stronger influence on generalization than mere label frequency.



(a) GDRO accuracies across domains (b) GDRO domain weights over training

Figure 3. Performance and weighting behavior of the GDRO algorithm ($\eta = 0.01$). (a) shows domain-wise accuracies during training, while (b) illustrates the corresponding domain weights.

Secondly, when rare classes are present, all adaptation methods perform significantly worse on those categories. As shown in Table 2, the *giraffe* class, which has only three target-domain training samples, experiences a severe drop in F1 across all models (0.3333 for DANN, 0.2105 for CDAN, and 0.1111 for DAN). This demonstrates that even with alignment-based adaptation, extreme class scarcity remains a dominant limiting factor for target-domain generalization.

Finally, when a class is entirely absent from the target domain, as in the *label-removed* condition (Table ??), the performance on other classes remains largely stable or even improves slightly compared to the *normal* setting. This indicates that alignment-based adaptation can still learn transferable representations for shared classes, even when one category is completely missing from the target domain distribution.

3.1.2. QUALITATIVE RESULTS

Figures 4 and 5, present two-dimensional t-SNE visualizations of feature embeddings for the DAN, DANN, and CDAN models, respectively. Each figure includes three subplots corresponding to the target-domain conditions, (a) Normal, (b) Concept Shift, and (c) Label Frequency Shift. Ideally, well-separated clusters would indicate effective class discrimination and domain alignment. However, the visualizations reveal partial overlap between several classes. In particular, all models exhibit notable confusion for the house category, with DANN showing the greatest degree of inter-class mixing.

3.2. Comparison of Domain Generalization Algorithms Against the ERM Baseline

Each model’s accuracy was evaluated on the target Sketch domain as well as on each source domain. Model selection was performed using the best validation accuracy criterion. The detailed results are presented in Table 3.

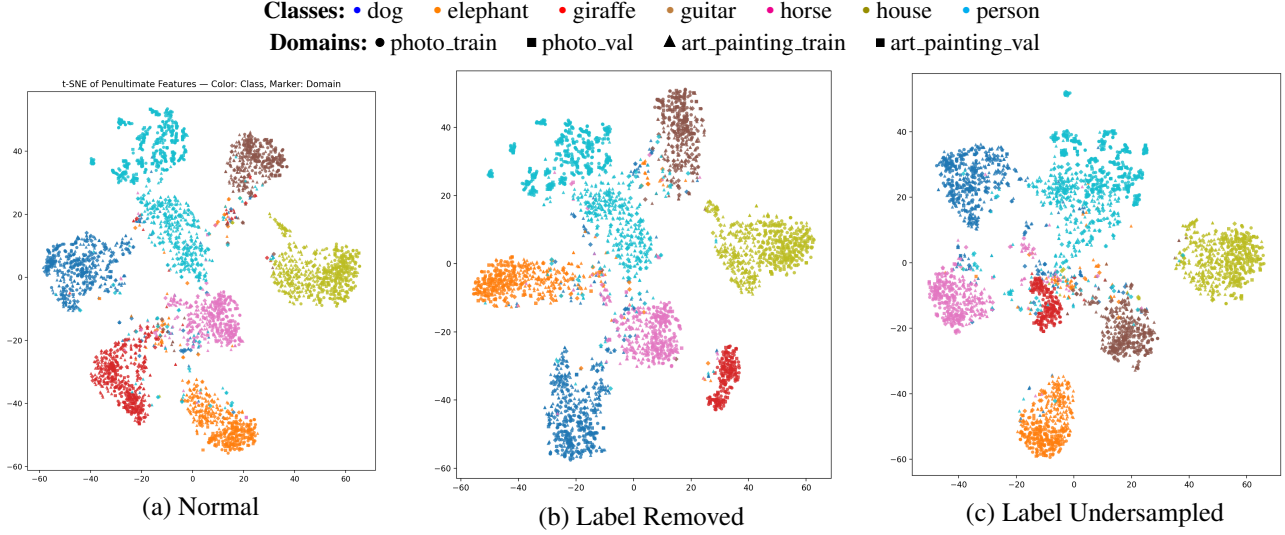


Figure 4. t-SNE visualizations for the DAN model under three target-domain conditions: Normal, Label Removed, and Label Undersampled. Legends above indicate class (color) and domain (marker).

ERM baseline: ERM attains a target (Sketch) accuracy of 72.74% with a high average source accuracy of 96.85%.

IRM: Throughout all algorithms IRM with $\lambda = 10$ achieves the highest target accuracy of 74.90% with a source average of 97.32% and a weighted penalty of 1.93×10^{-8} , which is very close to 0. The $\lambda = 0.1$ configuration follows closely at 74.52% on the target with a lower source average (95.92%) and penalty of 1.31×10^{-3} . Larger penalties i.e 100, 10000 collapse, yielding very low accuracies across both target and source domains (e.g., target 5.86% and 7.64%, respectively). The evolution of the IRM penalty during training is shown in Fig. 2. This shows the models optimization to drive these losses to zero, which can often impact their performance since the better performing models were in the earlier stages of training where the penalty was still higher, once again highlighting that IRM can be hard to optimize. Under weight perturbations, the IRM models exhibit consistent performance degradation across domains all domains. For the tests to measure cross-domain flatness, on average, the mean loss increased by approximately 0.086 and 0.051 across domains for both $\lambda = 0.1$ and $\lambda = 10$ respectively, and this increase was consistent across all domains, indicating that the model optimized within a region where perturbations affected all domains similarly, indicating perhaps a solution that is in the direction of invariance.

GroupDRO: After the IRM models, GroupDRO reaches the highest target accuracies up to 73.38% ($\eta = 0.01$) with strong source-domain performance reaching a maximum of 97.34%. This is closely followed by $\eta = 0.1$ and then with a larger difference $\eta = 0.001$. Across runs, *Photo* is typically the worst-performing source domain; correspondingly, it receives the largest training weight as training proceeds

(Fig. 3b), and its accuracy increases over training (Fig. 3a). Furthermore, the model criteria selections earlier checkpoints (1000-2000 steps), which highlights the requirement for strong regularization to see improvements over ERM for GDRO.

SAM: For $\rho = \{0.01, 0.05, 0.1\}$, SAM consistently results in target accuracies in the 70% range (e.g., 70.96%, 73.30%, 72.44% respectively) in Table 3), while maintaining strong source-domain accuracies (e.g., 96.94%, 96.86%, 96.04%) and even outperforming ERM on both fronts. To evaluate the flatness of the converged models, under controlled gradient-based weight perturbations, SAM exhibits order of magnitudes smaller increase in loss than ERM. (Fig. 1a). This is qualitatively supported by the flatter regions of the loss surface for SAM compared to ERM in the 2D subspace (Fig. 1c).

Since all the best models for each algorithms have target accuracies in the 70% range and source accuracies in the higher 90% range, our results show that for properly trained models, Domain Generalization algorithms come close to or outperform ERM results only by a very small margin.

3.3. CLIP

3.3.1. ZERO-SHOT AND FINE-TUNED LINEAR PROBES

The experiments revealed a trade-off between zero-shot generalization and domain-specific adaptation. CLIP’s pre-trained representations remained highly competitive even against fine-tuned linear probes trained with full supervision.

Zero-shot CLIP achieved strong cross-domain performance:

photo 99.70%, cartoon 97.65%, art painting 95.56%, and sketch 85.16%. The photo domain, aligned with CLIP’s pre-training distribution, achieved near-perfect accuracy, while sketch was most challenging due to minimal texture and color. The mean accuracy was 94.52%, with worst-group accuracy of 85.16%, forming the fine-tuning baseline.

Linear probes showed stable training dynamics. Initial losses (1.86–1.89) and accuracies (59–78%) improved to losses of 0.42–0.45 and accuracies of 94.09–97.64% by epoch 50. The probe excluding sketch reached the highest source accuracy (97.64%), while that excluding art painting reached 94.61%. On source domains, accuracies exceeded 90% across all: photo 99.70–99.82%, art painting 94.97–96.24%, cartoon 97.99–98.51%, and sketch 90.38–90.94%.

On target (held-out) domains, fine-tuning slightly reduced generalization: photo 99.64% (0.06), art painting 94.04% (1.51), cartoon 96.84% (0.81), and sketch 84.09% (1.07), averaging 0.86 points. Overall, mean accuracy improved from 94.52% to 95.46%, while worst-group accuracy rose from 85.16% to 89.02%, a 26% error reduction on sketch.

3.3.2. PROMPT LEARNING STRATEGIES

Prompt learning consistently improved target performance with distinct adaptation behaviors.

Zero-shot CLIP achieved 84.83% on sketch (MSP: 0.8825, entropy: 0.3424) and 99.70% on photo (MSP: 0.9846, entropy: 0.0734). Source-only prompt tuning (10 epochs) reduced loss from 0.15 to 0.0069 and raised source accuracy to 99.94%. Target accuracy improved to 85.93% (+1.10), with MSP 0.9094 and entropy 0.2588. Prompt drift was 3.4% (cosine similarity 0.9657 ± 0.0061), indicating mild adaptation. Adding target entropy minimization (15 epochs) increased adaptation. Loss decreased from 0.0604 to 0.0468; source accuracy remained 99.94%, and target accuracy reached 86.08% (+1.25). MSP rose to 0.9630, entropy dropped to 0.1024 (70%), and prompt drift increased to 12.9% (cosine 0.8711 ± 0.0494).

The source + target pseudo-labeling strategy (15 epochs) achieved the best results. Loss dropped from 0.0160 to 0.0118, with source accuracy 99.94% and target accuracy 89.90% (+5.06). MSP was 0.9451, entropy 0.1534, and prompt drift 6.3% (cosine 0.9374 ± 0.0208), reflecting strong yet aligned adaptation.

3.3.3. GRADIENT CONFLICT ANALYSIS

Gradient analysis showed cyclic alignment between domains, alternating between cooperation and conflict. At epoch 0, gradients from art painting and photo were weakly aligned (cosine 0.1881, 79.2° ; $\|g_{art}\| = 0.1108$, $\|g_{photo}\| = 0.1435$). By epoch 5, alignment fell to 0.0613

(86.5°), with diverging magnitudes. At epoch 10, alignment peaked at 0.6541 (49.2°), marking maximal cooperation. Severe conflict appeared at epoch 15 (0.7535, 138.9°), dominated by photo gradients. Conflict persisted at epoch 20 (0.2342, 103.5°), then partially realigned by epoch 25 (0.1242, 82.9°). Training concluded (epoch 30) with weak alignment and persistent magnitude asymmetry favoring photo. This cyclic pattern, alignment, divergence, cooperation, opposition, recovery, illustrates the non-monotonic optimization of multi-domain prompt learning.

3.3.4. CLOSED-SET AND OPEN-SET RECOGNITION

Prompt tuning enhanced closed-set accuracy but reduced open-set robustness.

Zero-shot CLIP achieved 82.93% on seen classes (MSP: 0.7487 ± 0.1893 , entropy: 0.7255 ± 0.3636) and 100% on unseen classes (MSP: 0.9415 ± 0.0487 , entropy: 0.2077 ± 0.1004). After 10 epochs of tuning, loss decreased from 0.3570 to 0.2551, and seen-class accuracy rose to 90.11% (+7.18), with MSP 0.8772 and entropy 0.3512. In open-set evaluation using tuned prompts for seen and zero-shot prompts for unseen classes, unseen accuracy dropped to (34.6%), with MSP 0.5830 and entropy 1.1568. Among 240 unseen samples, 83 were misclassified. OOD evaluation showed MSP 0.5537 ± 0.1456 (vs. 0.8772 in-distribution) and entropy 1.1531 ± 0.2470 ($3.28 \times$ higher). MSP-based OOD detection achieved AUROC 0.9059, FPR@95TPR 0.2385; entropy-based detection achieved AUROC 0.9397, FPR@95TPR 0.2114, indicating better entropy calibration. Prompt similarities ranged 0.6935–0.8015 (mean 0.7355 ± 0.0386), below the 0.85 threshold, confirming class-specific specialization. Cross-domain tuning on photo reduced loss from 0.0652 to 0.0306; sketch–photo prompt similarities averaged 0.7233 ± 0.0414 , comparable to within-domain drift (0.7355), confirming domain-specialized rather than shared representations.

4. Discussion

4.1. Domain Adaptation

The phenomenon of domain shift is well-documented in literature. (Li et al., 2017) introduced the PACS benchmark, showing that cross-domain performance drops are substantially higher when transferring from natural to stylized domains such as *art painting*, *cartoon*, and *sketch*. They quantified this through KL divergence between feature distributions and reported an average generalization gap of approximately 20% which is double that observed on earlier photo-only benchmarks. This clearly illustrates that large divergences between source and target distributions severely degrade the performance of models trained using ERM, which assumes identical data distributions during

training and inference.

In order to interpret this degradation, we examine the intrinsic differences between the *photo* and *art painting*. (Van Zuijlen et al., 2021) demonstrated that paintings, being synthetic in nature, deviate from the statistics of natural image. For instance, artists systematically apply top-left lighting, patterns that are deliberate, uniform, and non-natural (Van Zuijlen et al., 2021). Such stylization leads to a distribution shift that directly impacts models trained on natural image priors. The problem is further increased by the well-established *texture bias* of ImageNet-pretrained CNNs. (Geirhos et al., 2022) showed that ResNet-50 relies predominantly on local texture cues rather than global shape information, with 77.9% of predictions following texture in texture–shape conflict tests. This explains the decrease in performance when trained on *art painting* as art has different texture than photo.

This decrease in performance aligns with the theoretical framework of (Ben-David et al., 2010), that showed that the target-domain error $\epsilon_T(h)$ is bounded by the source error $\epsilon_S(h)$, the divergence term $\frac{1}{2}d_{\mathcal{H}\Delta\mathcal{H}}(P_S, P_T)$, and the joint error λ . In our normal ERM model the feature representation for the two domains are dramatically different hence the divergence term increases which increases the error.

On the contrary, our adaptation models that explicitly minimized the divergence component achieved higher accuracy on the *art-painting* domain relative to the ERM baseline, validating the expected behavior of the bound. However, there is a tradeoff with this improvement. Models that naively try to match the distributions often do so at the cost of discriminative features. This phenomenon is observed in our experiments as well. Our DANN model confuses the *person* class in the *art* domain with every other class in the *photo* domain (see Appendix A). This is in line with the literature (Xiao & Zhang, 2021), which reports that strong adversarial alignment can degrade class separability by forcing features from distinct categories to overlap in the shared representation space. As alignment strength increases, within-class compactness diminishes and inter-class boundaries collapse, resulting in mixed or entangled clusters in the feature space.

This happens because DANN performs marginal alignment, it aligns global feature distributions across domains without considering class boundaries. As a result, features from different categories that share similar textures or colors are pulled together, reducing between-class separability. In contrast, methods like DAN and CDAN incorporate additional structure into the alignment process: DAN aligns multi-layer representations more smoothly using kernel-based matching, while CDAN conditions the discriminator on class predictions to align features in a class-aware manner. Consequently, both retain better class structure, avoiding the over-alignment effect observed in DANN.

We also evaluated pseudo-labeling, which is frequently proposed as a simpler alternative for exploiting unlabeled target data. Pseudo-labeling can, in principle, improve target performance by providing direct supervision and bridging the absence of ground-truth labels. Yet in our experiments, it resulted in the opposite effect. The performance degraded sharply due to unreliable pseudo labels due to performance of our model on the target domain. Once incorrect pseudo labels entered the training loop, the model began to overfit on the training data and completely failing on evaluation as shown in Table 1. This failure mode aligns with findings from (Chhabra et al., 2024), which identify confirmation bias as the central risk. The paper shows that when pseudo labels are noisy, the model reinforces its own incorrect predictions, gradually drifting away from the true decision boundaries.

4.2. ERM or DG?

Our results show that ERM is at par with all other Domain Generalization training strategies. For PACS with Sketch as target, ERM attains competitive target performance (72.74%) while maintaining a high average on the source domains (96.85%; Table 3). This aligns with prior evidence that, under reasonable model capacity and training protocols, ERM can match or surpass many domain generalization (DG) methods when hyperparameters tuned fairly (Gulrajani & Lopez-Paz, 2020). In our setting, architectural regularization (frozen BatchNorm, dropout $p = 0.3$) and checkpoint selection by best validation likely further strengthened the ERM reference.

IRM with a moderate penalty achieves the best target accuracy: $\lambda=10$ reaches 74.90% on Sketch with strong source averages (97.32%), and improves the worst-source accuracy from ERM’s 95.11% (Photo) to 95.60% (Table 3). A smaller penalty ($\lambda=0.1$) yields a similar target result (74.52%) but reduces source average to 95.92%, consistent with trading off in-domain fit to encourage invariance. In contrast, large penalties ($\lambda \in \{100, 10,000\}$) collapse, driving accuracies to single digits on the target and markedly lower values on sources. These outcomes are consistent with IRM’s known optimization sensitivity: with too much penalty, the solution can learn trivial representations; with too little, the objective under-enforces invariance (Arjovsky et al., 2020). We show that IRM can slightly improve target performance relative to ERM when the penalty is tuned, but it remains hard to optimize requiring careful tuning and can underperform or collapse which is consistent with literature (Rosenfeld et al., 2021).

GroupDRO attains target accuracies in the low-70% (e.g., 73.38% at $\eta=0.01$ in Table 3) while consistently improving the worst-source accuracy. Training traces show the algorithm increases weight on the weakest source (Photo) and

its accuracy rises over training (Fig. 3). Best-checkpoint selections for GroupDRO tend to occur earlier, consistent with the need for stronger regularization and early stopping to avoid overfitting the worst-group objective (Bai et al., 2025).

Lasrly, SAM achieves flatter solutions with strong source and target accuracy. This is supported by literature that it can improve performance on both seen and unseen domains (Foret et al., 2021). Furthermore, Perturbation tests indicate SAM exhibits a smaller loss increase than ERM for comparable weight perturbation magnitudes. Different ρ did not impact performance too much, and 0.05 performed the best, which is supported by the original paper. However, a major problem with SAM is it requiring significantly more compute (requiring gradients to be calculated twice), but the gains are not as significant, which raises questions about whether it is practical.

4.3. Invariance versus discriminability in DG

For PACS, ERM remained competitive. Discriminative-based approaches like IRM and GroupDRO only gave marginal gains that too requiring specific hyperparameter. These patterns suggest that, for this dataset, enforcing discriminability is the more suitable approach since it is easier to optimize and still achieves competitive performance. The results indicate each ingredient helps different aspects (IRM/GDRO for invariance and worst-group robustness; SAM for flatness and stability), but none universally dominates. A promising direction might be to combine SAM with invariance objectives, and selecting checkpoints by worst-group or domain-held-out criteria—so that the learned minima are both flat and domain invariant.

4.4. CLIP

4.4.1. ZERO-SHOT GENERALIZATION OF CLIP ACROSS DOMAINS

The zero-shot results demonstrate that CLIP’s pretrained representations generalize strongly across diverse visual styles. Its mean accuracy of 94.52% and worst-group accuracy of 85.16% confirm that multimodal training enables effective transfer even to domains far from its pretraining data. CLIP performs best on photo (99.70%), consistent with its internet-scale training distribution dominated by natural images. High performance on cartoon (97.65%) and art.painting (95.56%) further indicates that CLIP’s vision–language alignment preserves semantic consistency across stylized renderings.

The largest degradation appears in sketch (85.16%), where the absence of color and texture limits the cues leveraged by CLIP’s vision encoder. Despite this, achieving 85% accuracy without domain-specific training highlights the

robustness of its geometric and structural representations.

Fine-tuning revealed a clear adaptation–generalization trade-off. Linear probes trained on three source domains improved in-domain performance but slightly reduced accuracy on the held-out domain (average drop: -0.86 points). Fine-tuning encouraged reliance on domain-specific visual cues—such as color and texture, enhancing source accuracy but reducing transfer to unseen domains. For example, sketch accuracy decreased from 85.16% (zero-shot) to 84.09% (fine-tuned on photo, art, and cartoon). Conversely, photo performance remained stable ($99.70\% \rightarrow 99.64\%$), reflecting its proximity to CLIP’s pretraining distribution. Thus, domains visually aligned with CLIP’s training data resist overfitting, while stylized domains degrade under adaptation.

Although fine-tuning reduced target generalization, it increased average accuracy ($94.52\% \rightarrow 95.46\%$) and worst-group accuracy ($85.16\% \rightarrow 89.02\%$), as in-domain gains outweighed losses on the held-out target. This improvement reflects specialization rather than enhanced robustness, indicating that fine-tuning benefits scenarios with known deployment domains but reduces performance on unseen ones.

4.4.2. PROMPT LEARNING FOR DOMAIN ADAPTATION

Prompt learning effectively balanced domain adaptation and semantic invariance. All strategies improved target (sketch) accuracy over the zero-shot baseline (84.83%), with pseudo-labeling achieving the best performance (89.90%). This demonstrates that lightweight text-space optimization can significantly enhance generalization under severe visual shifts such as photo-to-sketch transfer.

Prompt drift quantified adaptation intensity. Source-only tuning showed minimal drift (cosine = 0.9657, +1.10% gain), preserving pretrained semantics but underfitting the target. Entropy minimization induced stronger drift (0.8711) and overconfidence (entropy = 0.1024) with marginal gain (+1.25%), indicating overfitting. Pseudo-labeling achieved moderate drift (0.9374), effectively capturing target structure while retaining alignment with pretrained semantics. This 6% drift achieved the best trade-off between adaptation and generalization.

Prompt learning offers three advantages over full-model fine-tuning: (1) high parameter efficiency with minimal risk of catastrophic forgetting, as only text embeddings are updated; (2) operation within CLIP’s contrastive text-embedding space, ensuring interpretable adaptation; and (3) drift regularization that preserves zero-shot alignment while allowing controlled domain-specific modification.

However, naive optimization remains unstable. Source-only tuning underfits due to lack of target feedback, while entropy minimization overfits through overconfident pseudo-labels.

Pseudo-labeling mitigates this by selectively incorporating reliable target samples, providing a self-supervised alignment signal that improves calibration and stability, consistent with prior self-training work (Lee, 2013; Zou et al., 2019).

4.4.3. GRADIENT CONFLICT AND ALIGNMENT IN MULTI-DOMAIN PROMPT LEARNING

Gradient analysis revealed frequent conflicts between art_painting and photo domains, reflecting competing optimization objectives. Across seven checkpoints, 34% of updates showed negative cosine similarity (destructive interference), 17% strong alignment, and the remainder weak or orthogonal, indicating unstable cooperation.

The strongest conflict occurred at epoch 15 ($\cos \theta = -0.7535$, 138.9°), where photo gradients ($\|g_{photo}\| = 0.1948$) emphasized texture cues, while art gradients ($\|g_{art}\| = 0.1015$) emphasized stylization. These opposing directions caused update cancellation and training stagnation. Early epochs evolved from weak alignment (epoch 0: 0.1881) to near-independence (epoch 5: 0.0613), followed by temporary cooperation at epoch 10 (0.6541) before collapsing into strong conflict at epoch 15.

This cyclic behavior, cooperation, divergence, opposition, and recovery, shows that photo gradients, typically 1.5–2× larger, dominate optimization, leading to oscillatory rather than convergent dynamics. Gradient alignment promotes updates beneficial to multiple domains by emphasizing shared high-level semantics (e.g., shape) over low-level cues (e.g., color). Misalignment stems from divergent texture and lighting preferences.

Methods such as PCGrad or GradCos can mitigate destructive interference by projecting gradients onto mutually non-conflicting directions. At epoch 15, orthogonal projection of the photo gradient relative to the art gradient would prevent cancellation, enabling joint improvement and stability under large domain shifts. Enforcing gradient alignment thus constrains optimization toward invariant, domain-agnostic representations.

4.4.4. IMPACT OF PROMPT TUNING ON OPEN-SET RECOGNITION

Prompt tuning on fixed classes improved closed-set accuracy but impaired open-set recognition. Seen-class accuracy increased from 82.93% to 90.11%, while unseen-class accuracy dropped from 100% to 65.42%. This indicates that tuning deforms CLIP’s embedding space, pulling unseen concepts toward tuned-class regions and distorting global geometry.

While tuned prompts partially reflected uncertainty, unseen confidence decreased from 0.8772 (seen) to 0.5537, they re-

mained poorly calibrated. Entropy-based uncertainty proved more reliable: unseen samples exhibited 3.28× higher entropy, yielding AUROC 0.9397 versus 0.9059 for MSP-based detection, confirming entropy’s superior calibration.

Prompt embedding similarity further explains the performance drop. Tuned prompts had an average cosine similarity of 0.7355 with zero-shot versions (all < 0.85), showing substantial drift and over-specialization. This distortion disrupted CLIP’s contrastive semantic structure, reducing separability between known and novel classes.

Cross-domain analysis supports this view. Few-shot tuning on photo reduced loss from 0.0652 to 0.0306 but produced low similarity between photo and sketch prompts (0.7233), comparable to within-domain drift (0.7355). Thus, tuning primarily captured low-level, domain-specific cues, such as color and texture, rather than abstract semantics. The comparable intra- and inter-domain drift confirms that prompt tuning overfits to visual style, diminishing robustness to unseen classes and domains.

5. Conclusion

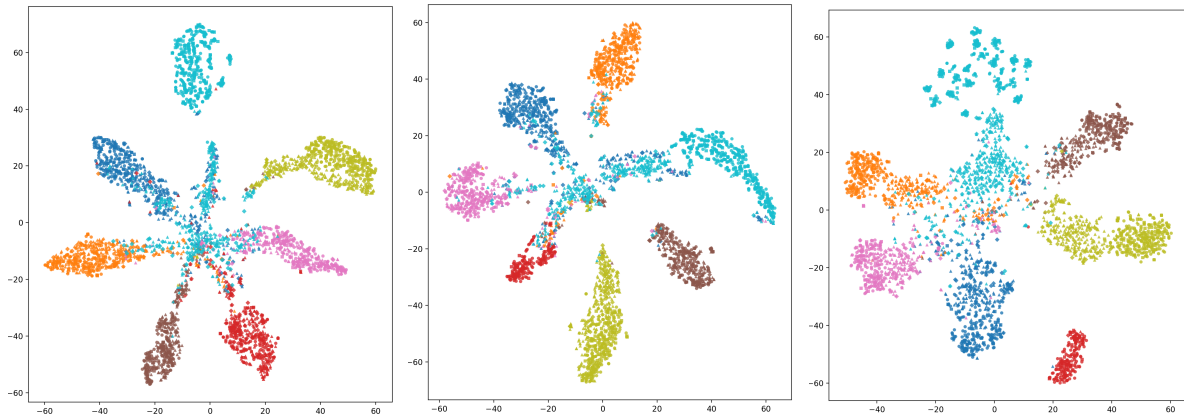
For domain adaptation, explicitly aligning source and target distributions improves transfer up to a point, but excessive alignment harms class separability. CDAN and DAN achieve comparable accuracy, whereas DANN underperforms due to over-alignment and feature collapse. Pseudo-labeling achieves optimal transfer with moderate drift, while aggressive entropy minimization produces overconfident predictions with minimal gains. For domain generalisation, ERM remains a strong baseline, often matching or surpassing DG methods when trained with proper regularization. IRM and GroupDRO can improve robustness but are sensitive to optimization, and SAM flattens solutions at high computational cost without significant performance gains. These findings highlight that balancing invariance and discriminability is crucial, with discriminative approaches being the most practical in typical settings. For prompt tuning in CLIP, fine-tuning improves seen-domain accuracy but degrades performance on unseen domains, while zero-shot CLIP retains strong cross-domain transfer. Gradient analysis reveals severe conflicts in multi-domain optimization, and open-set experiments show that prompt optimization alters embedding geometry, reducing recognition of unseen classes. These results emphasize that prompt tuning introduces overfitting, gradient instability, and embedding drift, suggesting that moderate, carefully controlled adaptations are necessary to preserve robustness and generalization.

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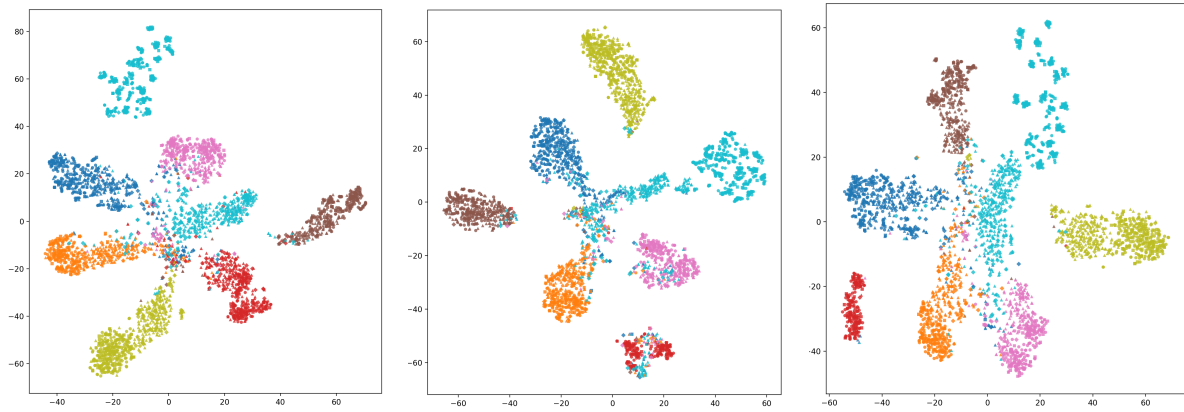
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A. Domain Adaptation Figures



(a) DANN: Normal, Label Removed, and Label Undersampled.



(b) CDAN: Normal, Label Removed, and Label Undersampled.

Figure 5. t-SNE visualizations for DANN and CDAN models under three target-domain conditions.